Unit Scales

SKU:U108













Description

Unit Scales is a 20 kg integrated weighing sensor unit with an I2C interface. It integrates an **STM32F030** microcontroller, acquisition circuit, and a 20 kg weighing sensor internally. The unit uses the **HX711** dedicated weighing acquisition chip, features a tare button, and a programmable **RGB** status indicator. Users can read weight data via the I2C interface. This unit provides a highly integrated weighing detection solution, suitable for applications such as quality inspection, piece counting, and object movement detection.

Features

- HX711:
  - High-precision 24-bit ADC
  - Programmable gain amplification of 32, 64, and 128
  - 10SPS output data rate
- I2C interface
- Development platforms: Arduino, UIFlow (Blockly, Python)

Includes

- 1 x Unit Scales

- 1 x HY2.0-4P Grove cable (20cm)

Applications

- Smart scales
- Kitchen scales
- Item counting devices
- Human presence detection

Specifications

Specification	Parameter
MCU	STM32F030F4P6
ADC	HX711
Communication Interface	I2C @0x26
LED	SK6812 (PA7 interface)
Button	PA2 interface
Housing Material	Plastic
Product Size	56.0 x 40.0 x 41.0mm
Product Weight	83.5g
Package Size	57.0 x 42.0 x 41.0mm
Gross Weight	92.3g

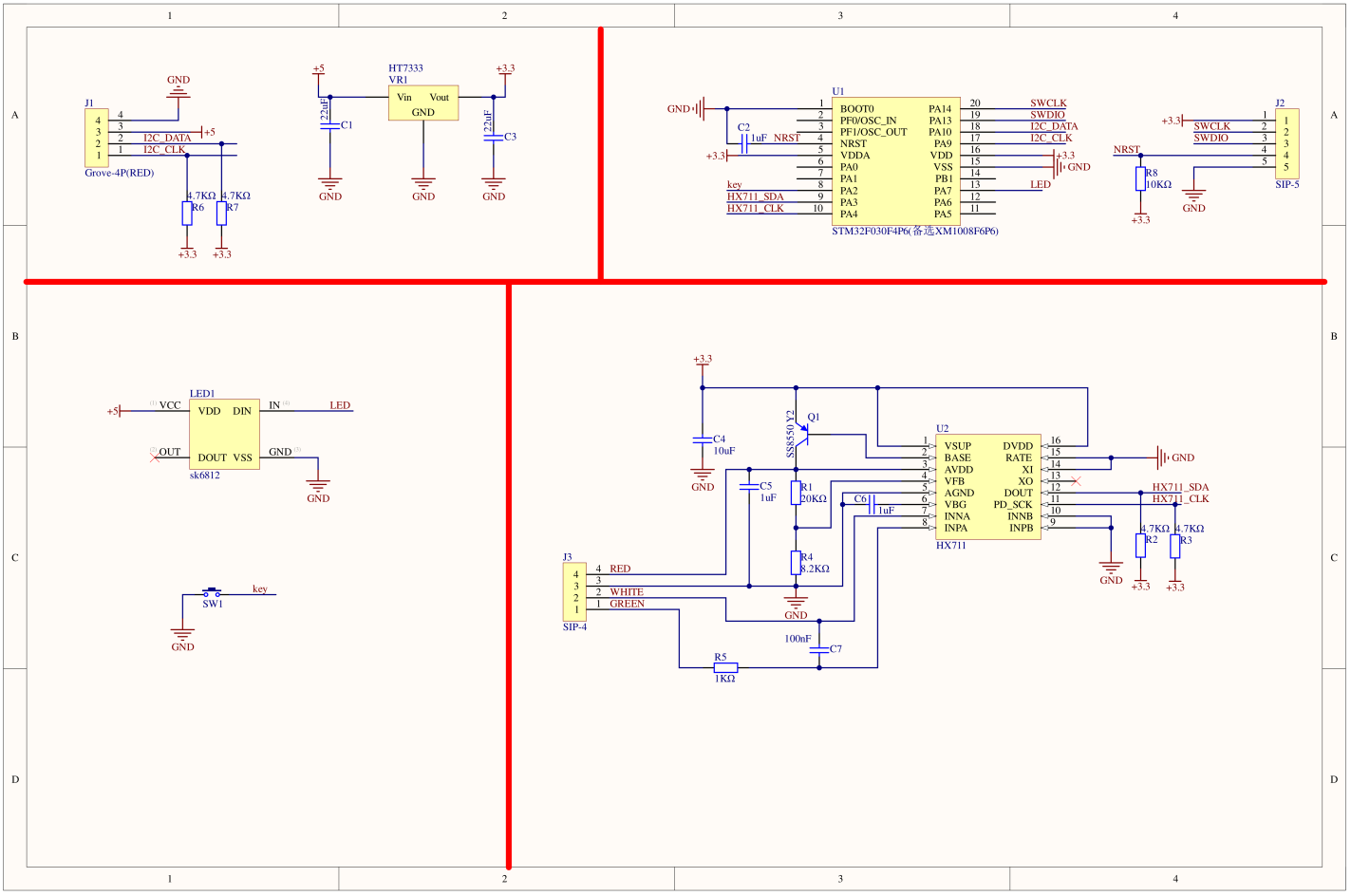
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Unit Scales Weighing Error Description

Weight (g)	50	100	250	500	1000
Absolute Error (g)	0.45	0.25	0.20	0.25	4.15
Relative Error	0.9%	0.3%	0.1%	0.1%	0.4%

Schematics

- Unit Scales Schematics PDF

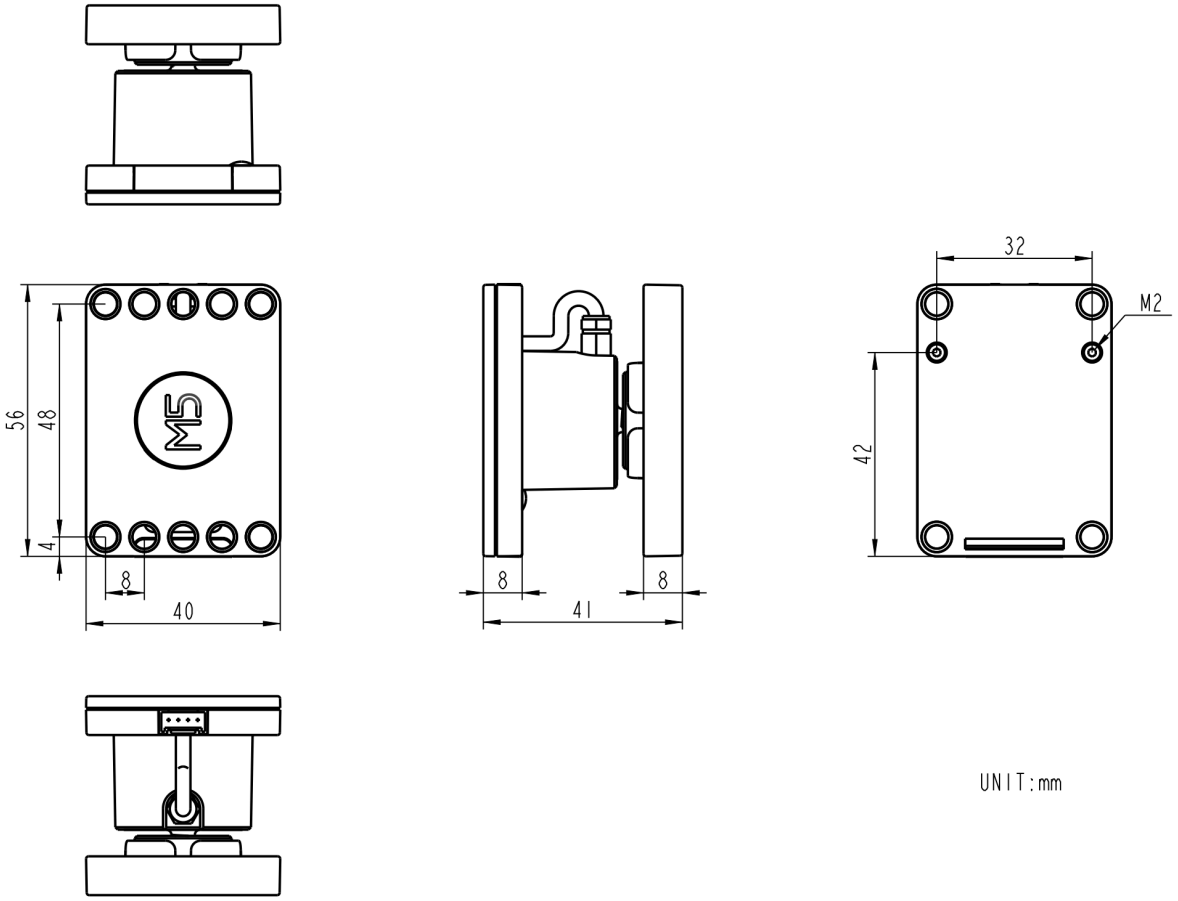


PinMap

Unit Scales

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit Scales Structure Files

Datasheets

- HX711

Softwares

Arduino

- Unit Scales Arduino Library
- Unit Scales Arduino Tutorial
- Scales Demo for Basic
- Scales Demo for ATOM

UiFlow1

- Unit Scales UiFlow1 Docs

UiFlow2

- Unit Scales UiFlow2 Docs

Internal Firmware

- Unit Scales Internal Firmware

Protocol

- Unit Scales I2C Protocol

EasyLoader

Easyloader	Download Link	Note
Unit Scales Test Easyloader	download	/

Video

Product Comparison

Product Compare					
	Unit Scales	**Scales Kit**	**Unit Mini Scales**	**Unit Weight**	**Unit Weight-I2C**
Communication Protocol	I2C	ADC	I2C	ADC	I2C
Chip Solution	STM32+HX711	HX711	STM32+HX711	HX711	STM32+HX711
Sensor Range	0 ~ 20kg	0 ~ 200kg	0 ~ 5kg	Depends on the strain gauge and controller chip	Depends on the strain gauge and controller chip